

Dynamic Versus Static Analysis: Is the LSA 2027 Summer Institute Guiding Linguistics in the Wrong Direction?



[MANOVA AI](#)

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The LSA Executive Committee is delighted to announce that the 2027 Linguistic Institute will be returning to the University of Oregon, building on the success of the 2025 Institute held there. The Co-directors will be Volya Kapatsinski (Chair of Linguistics), Kaori Idemaru (Chair of East Asian Languages and Literatures), and Spike Gildea (Professor of Linguistics).

The Co-directors explained, "Language is not static but continually evolving. Based on our [previous] experience, we propose a focus on dynamics across the timescales of processing, learning, and change."

Coincidentally, I recently posted a short text on dynamic and static systems, *Frequency versus Productivity and Why AI Does Not Use the Latter*

(<https://lingbuzz.net/lingbuzz/010164>), as part of the series *The AI Perspective on Linguistic Theory*. In this text, I argue that treating language as a dynamic system creates an open space for alternative linguistic DESCRIPTIONS, but it does not lead to fluent language GENERATION. Moreover, irregular systems such as human language can be generated only if treated as closed (static).

Thus, computer scientists solved the problem of fluent language generation after reformulating language as a settled (static, or closed) system based not on an open set

of words but on a closed set of tokens (the term *token* does not have the same meaning in computer science as it does in linguistics). This change of approach is what made modern LLMs possible (ChatGPT was released in November 2022), and the advantages of the closed-system approach are now clearly visible: major technological breakthroughs, numerous practical applications, and billions invested in AI. At the same time, linguists have continued to devote their efforts to describing language as a dynamic system: historical development, variation within and across languages, and so on, i.e., seeing language as a system full of change and language-specific irregularities that calls for permanent reanalysis.

To be sure, generative grammar also attempted a closed, formal analysis. However, it struggled with generation because it assumed that a small set of recursive rules could project an infinite, perfectly rule-governed space—an approach that breaks down against the inherent irregularities of natural language, to say nothing of its reliance on inverted syntactic trees that grow from leaves to root, see *Modeling Language Without Language: A ChatGPT Lesson for Language Research* (<https://ling.auf.net/lingbuzz/008998>).

Modern AI solved the computational problem of fluent language generation in a way that formal generative grammar did not. At the moment of execution, every functional system, whether a human brain or a graphics processing unit (GPU), operates on a closed, bounded state. At time T_0 , the executor is not computing over an infinite set of element combinations but over a specific, bounded physical network of neurons and stored memory; cf. the transformer architecture in AI models and the step-by-step selection of tokens. Thus, in an exact slice of time, the system is always closed (settled, static, or bounded). This distinction between the state of a system at time T and how that system behaves over time, at the next point, $T + 1$, actually explains why AI and the human brain succeed in language generation while linguistic theory does not. Fluent generation becomes computationally manageable because each step takes place within a settled, bounded system.

I am aware that most linguists are not used to thinking of language processing as a sequence of discrete computational states and might therefore ask how language could ever be treated as a finite system. Yet linguistics already has subfields built precisely on the idea of a finite, discrete system of elements: phonology and graphematics.

Phonology and graphematics both rely on strictly bounded sets of discrete elements and features to generate well-formed phonological and graphematic structures through settled combinatorial rules. Although they are not formulated in terms of computational states, they demonstrate that treating a level of language as a finite, bounded system is already a well-established linguistic practice. In this sense, phonology and graphematics can be viewed as oversimplified precursors to an LLM. Broadening this logic, the internal architecture of modern AI models suggests cross-linguistic solutions that can scale such finite, structural organization across all levels of language—revealing that linguistics and AI are, in fact, remarkably close.

Of course, in which direction to push linguistic research (because this is what the 2027 LSA Summer Institute will do) depends on what linguistics wants to achieve as a field. As mentioned above, a DYNAMIC approach yields DESCRIPTION, while a STATIC one yields GENERATION. Perhaps it is already time for linguists' understanding and analysis of language to change—at least to the extent that AI has changed language generation.

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